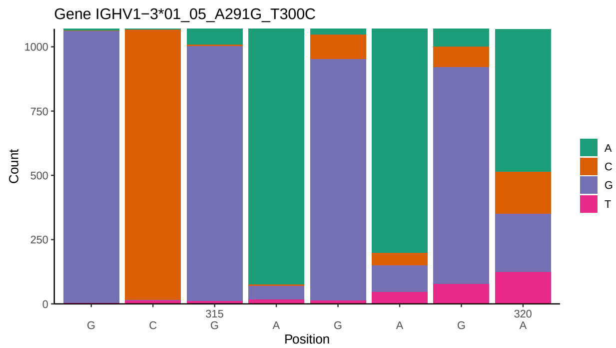
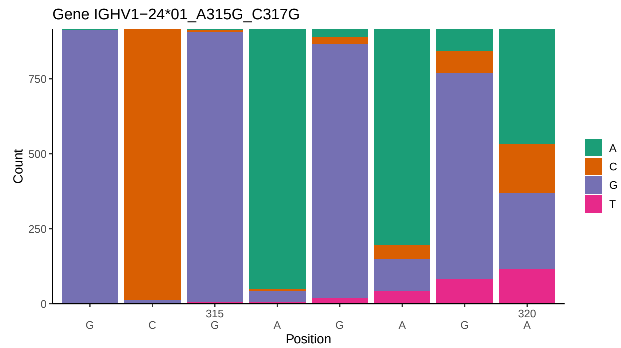
OGRDBstats Report

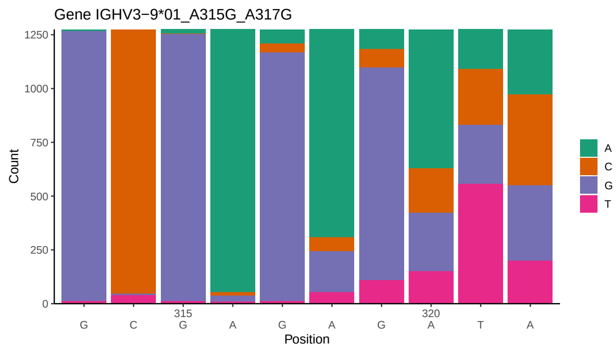
Contents

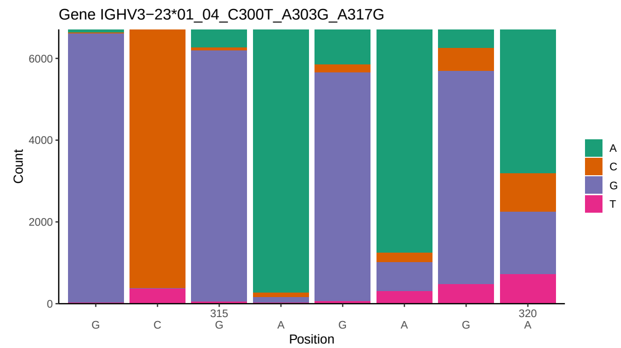
1 Novel sequence analysis

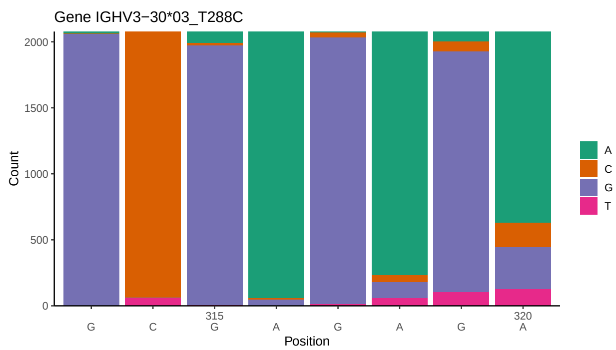
1.1 End-nucleotide composition

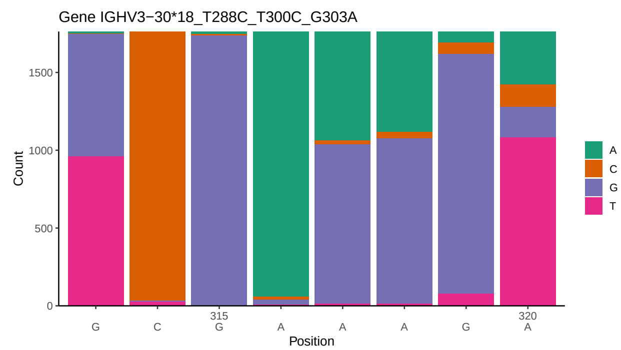


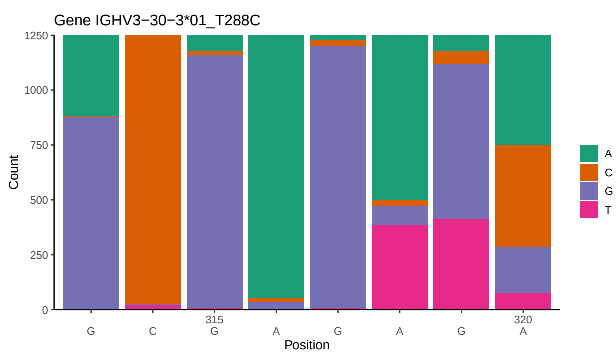


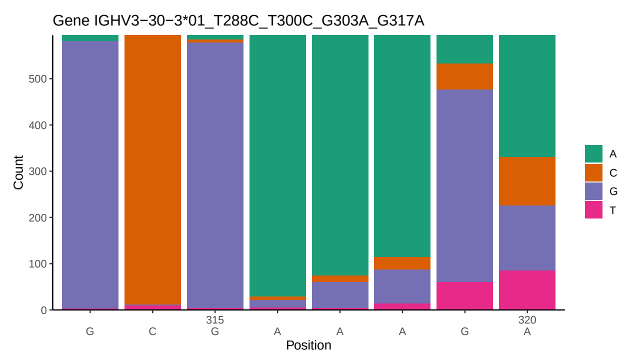


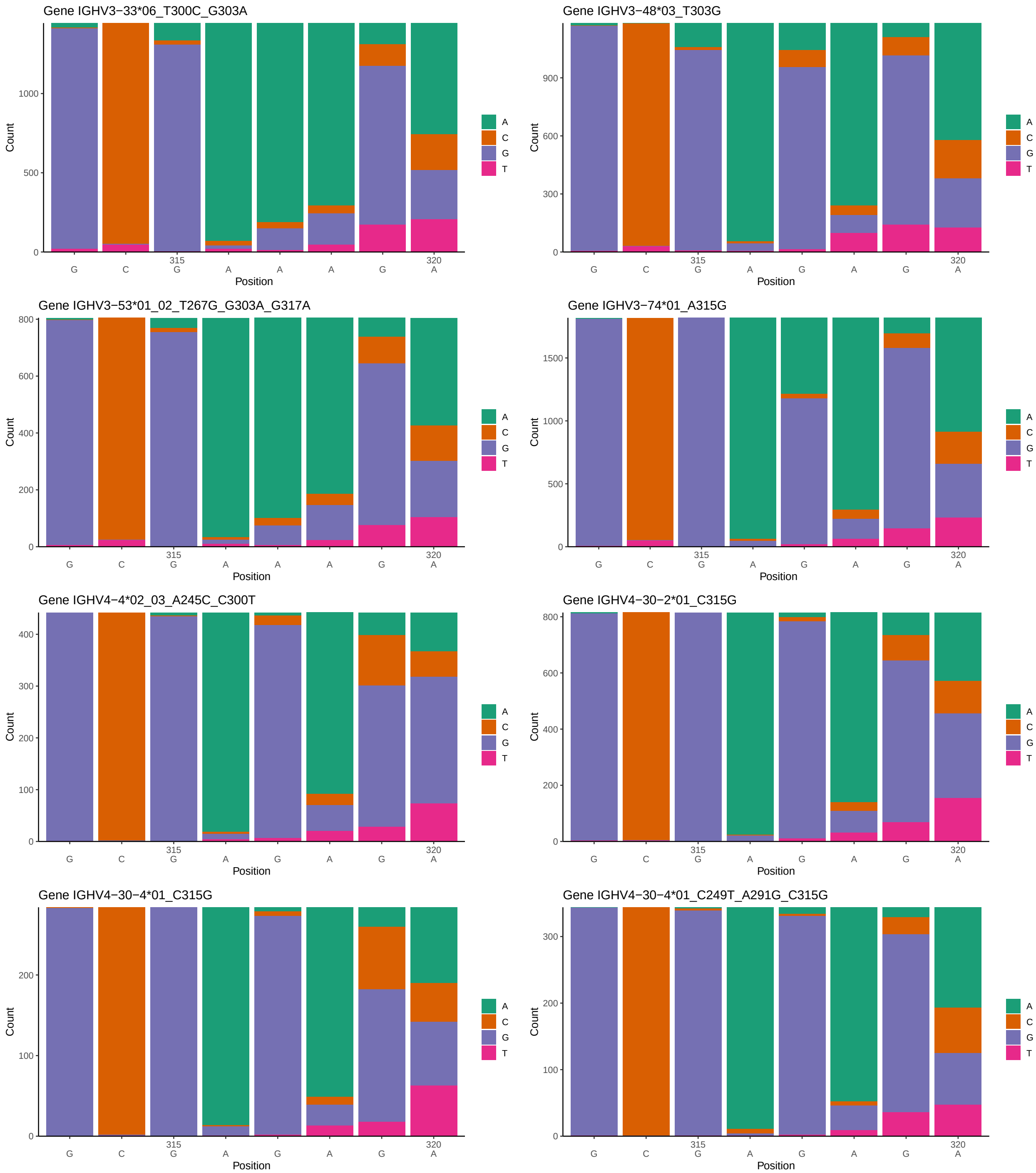


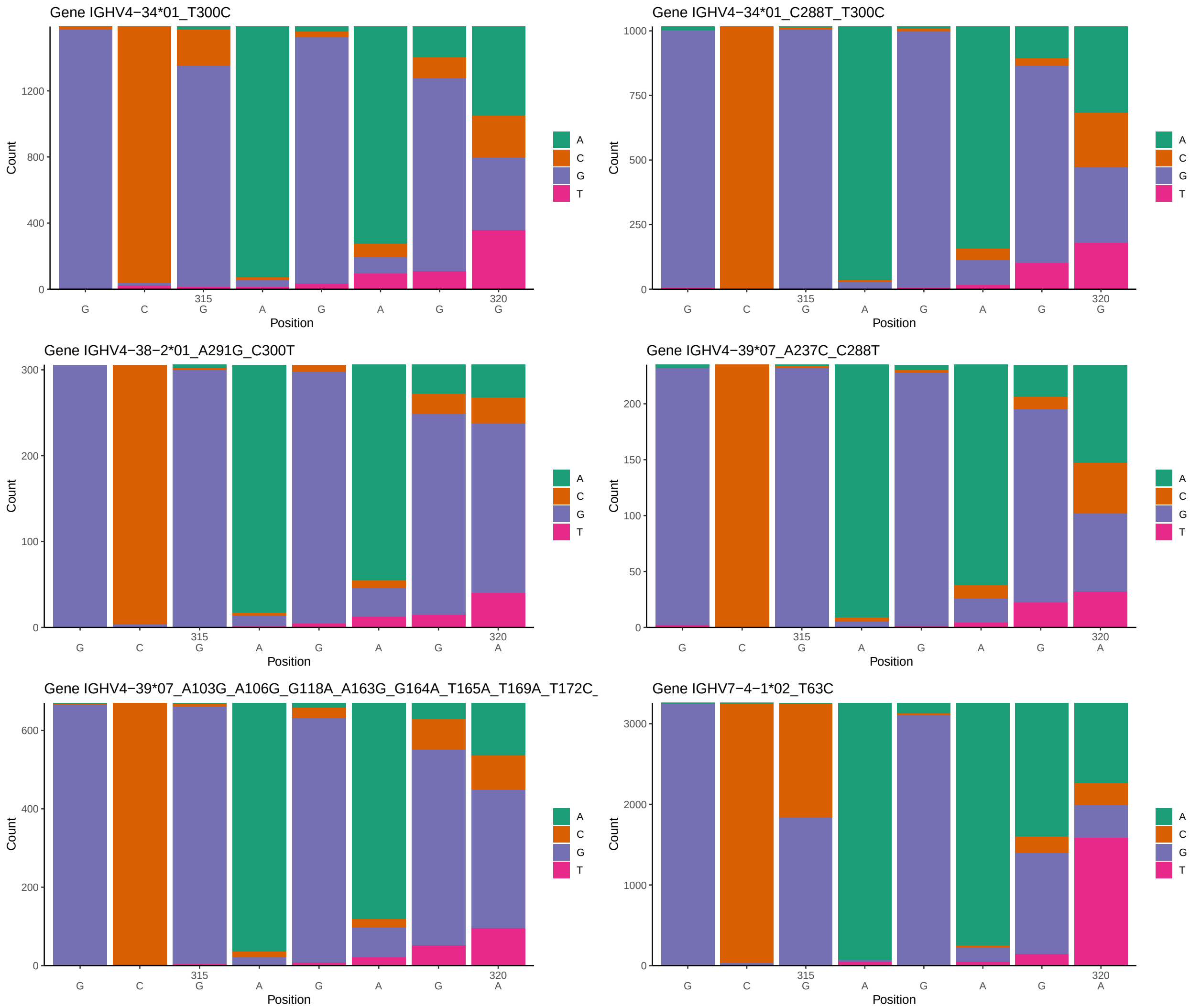




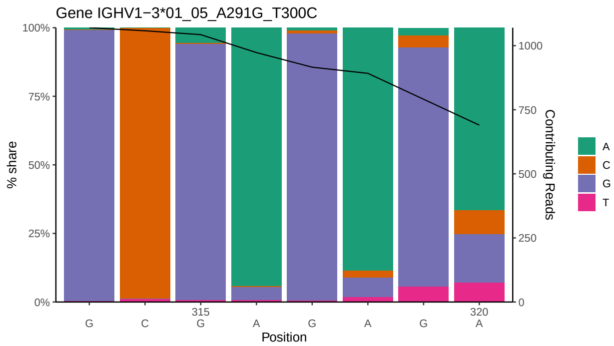


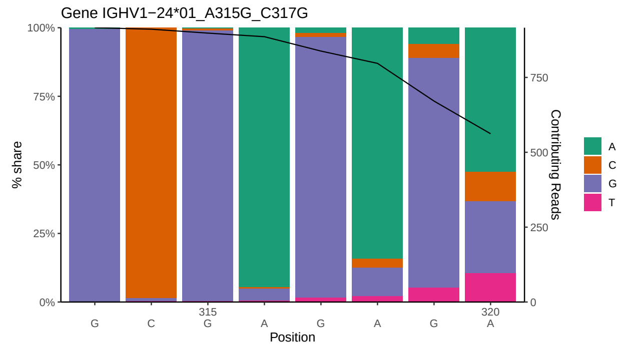


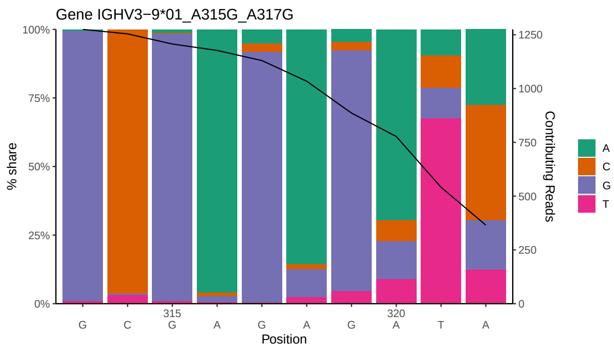

Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV3−53*01_02_T267G_G303A_G317A
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−30−4*01_C249T_A291G_C315G
Count
Position
G C G A A A G A
315
320
A
C
G
T

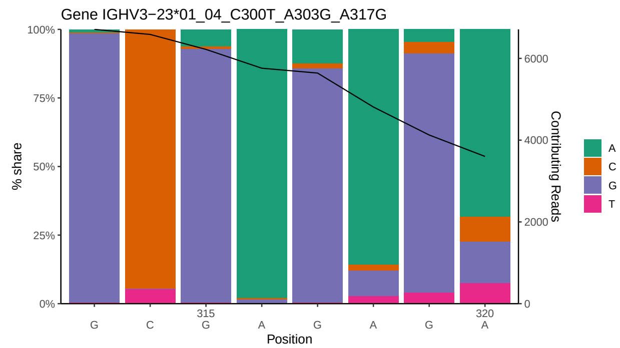

Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−38−2*01_A291G_C300T
Gene IGHV4−39*07_A237C_C288T
Gene IGHV4−39*07_A103G_A106G_G118A_A163G_G164A_T165A_T169A_T172C_
Gene IGHV7−4−1*02_T63C
Count
Position
A
C
G
T

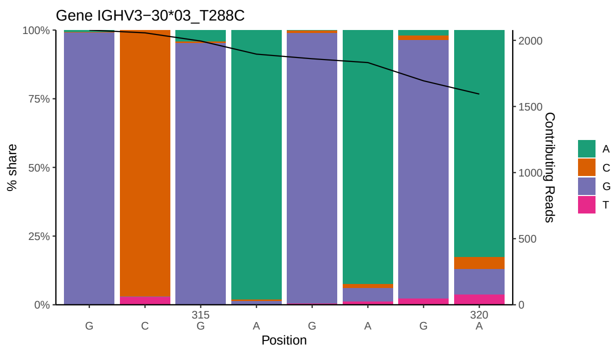
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

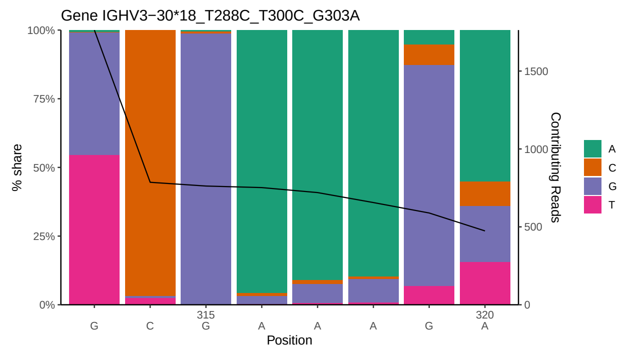


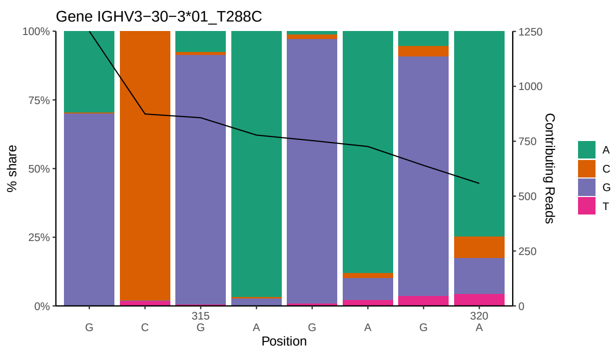


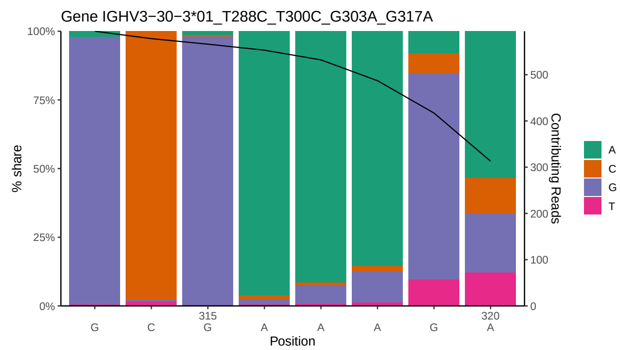


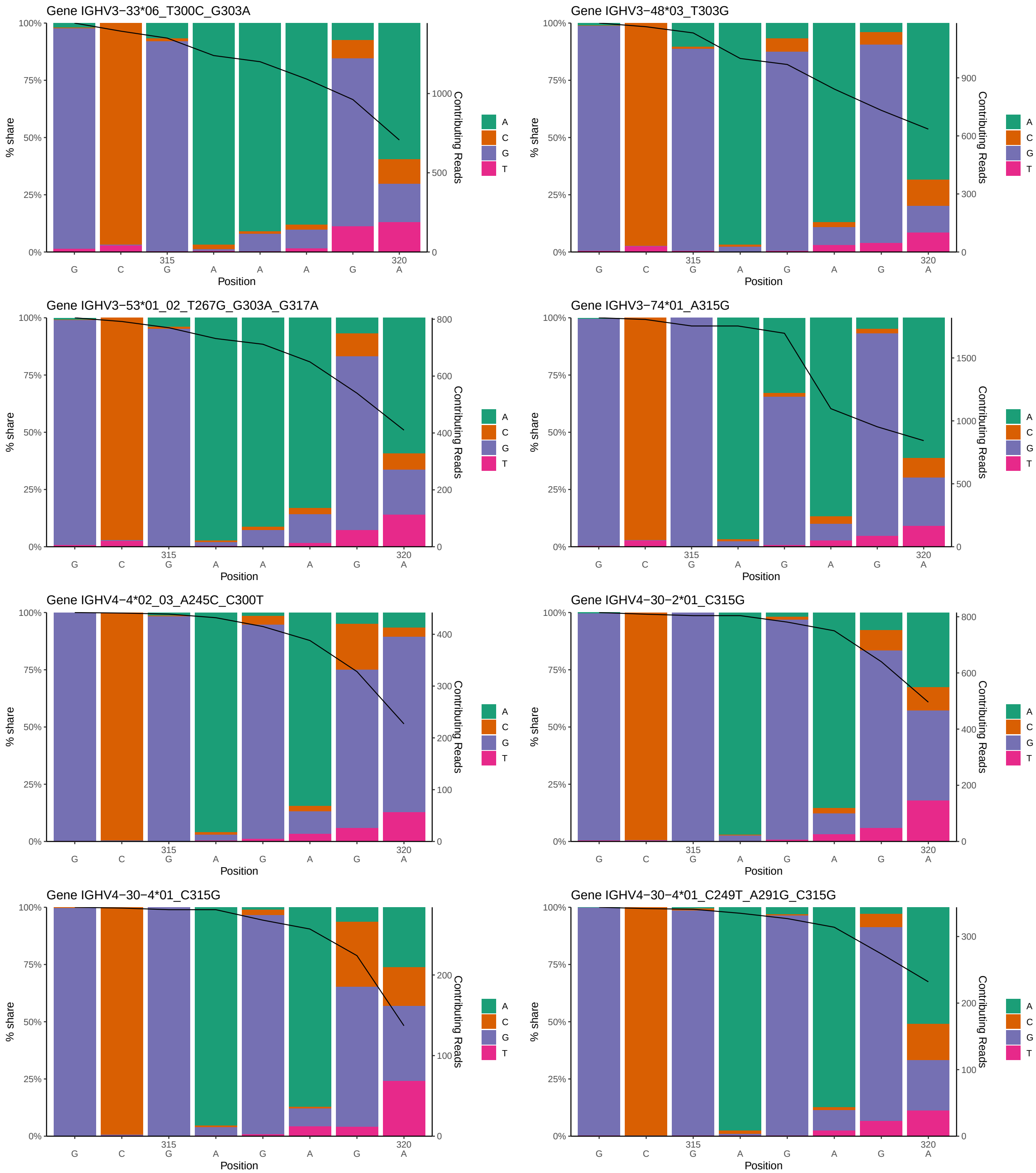


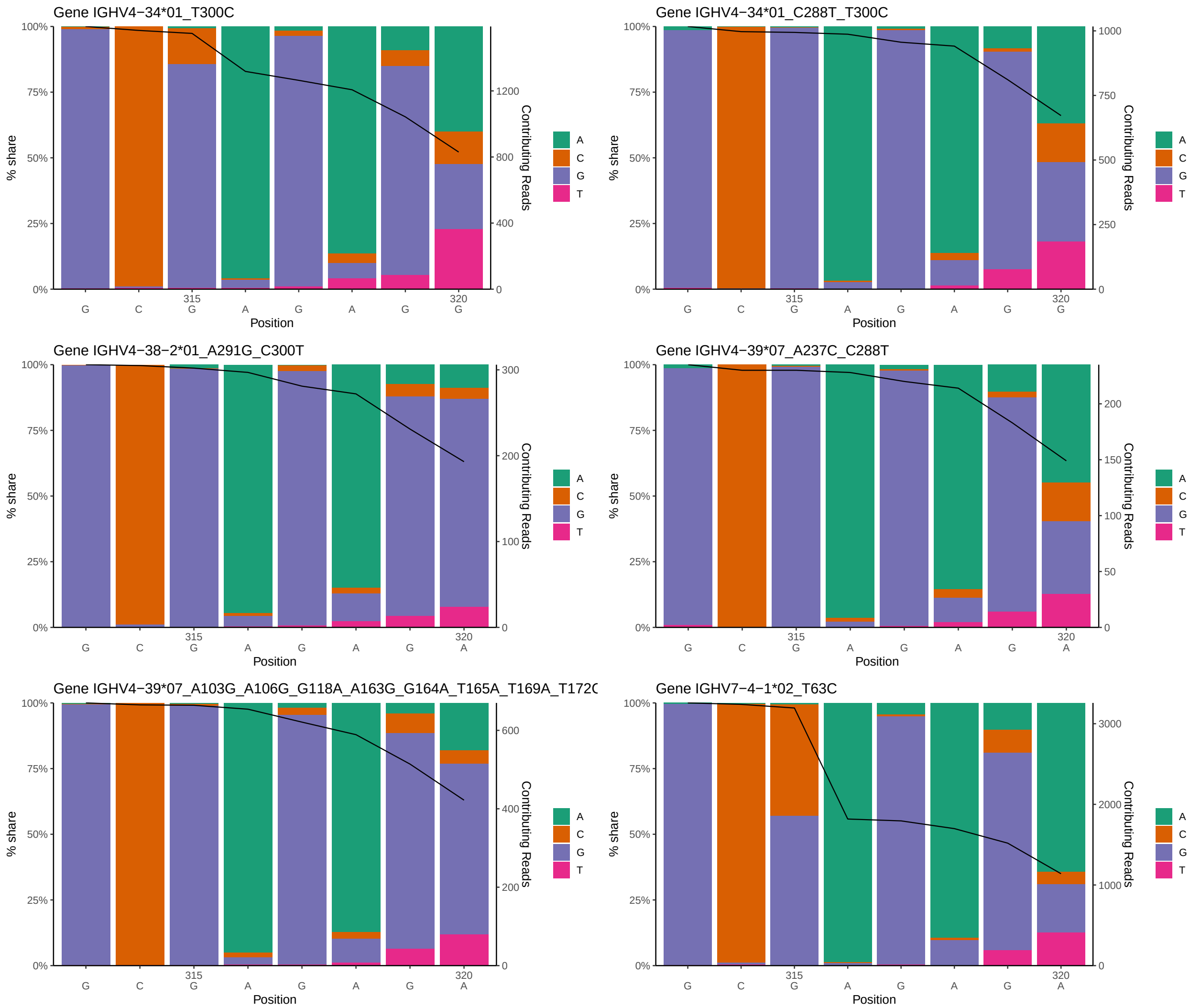




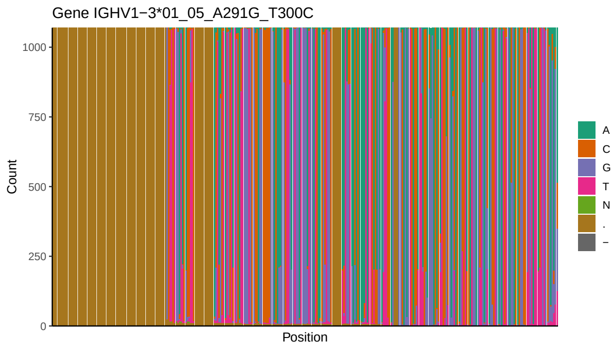


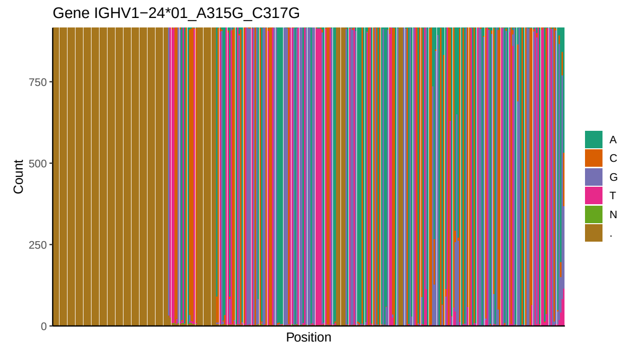


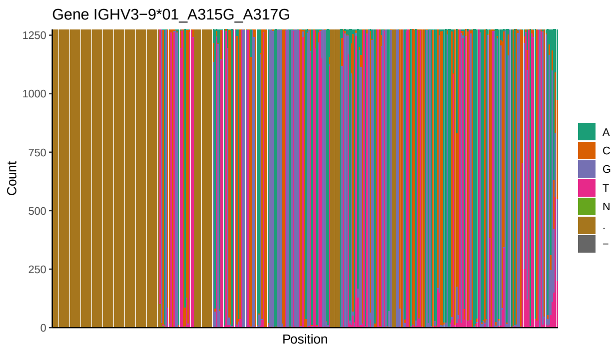

Gene IGHV3−33*06_T300C_G303A
Gene IGHV3−48*03_T303G
Gene IGHV3−53*01_02_T267G_G303A_G317A
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−30−4*01_C249T_A291G_C315G
% share
Contributing Reads
Position
0%
25%
50%
75%
100%
315
320
G
C
G
A
A
A
G
A
A
C
G
T

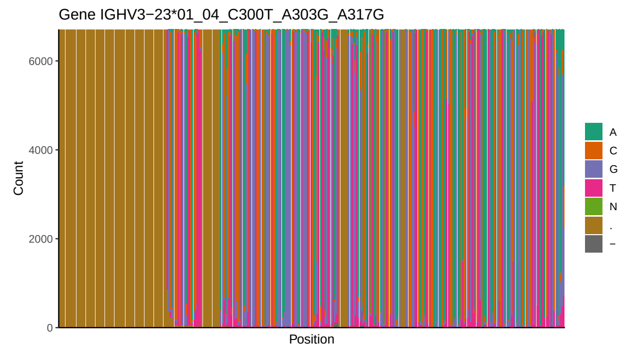

Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−38−2*01_A291G_C300T
Gene IGHV4−39*07_A237C_C288T
Gene IGHV4−39*07_A103G_A106G_G118A_A163G_G164A_T165A_T169A_T172
Gene IGHV7−4−1*02_T63C
% share
Contributing Reads
Position
A
C
G
T

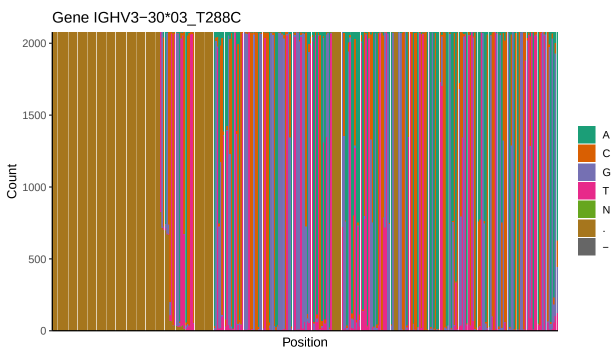
1.3 Whole-sequence composition of each assigned read

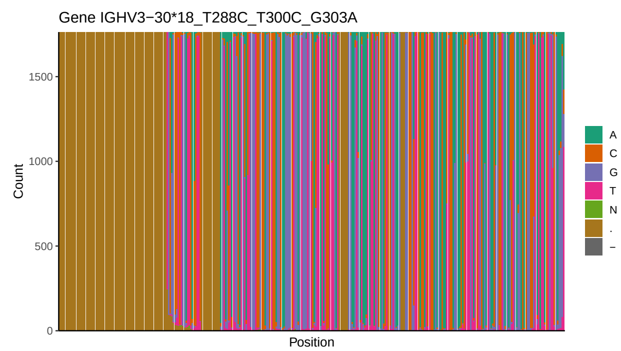


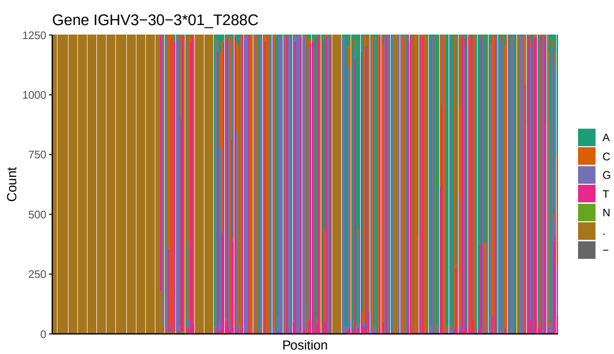


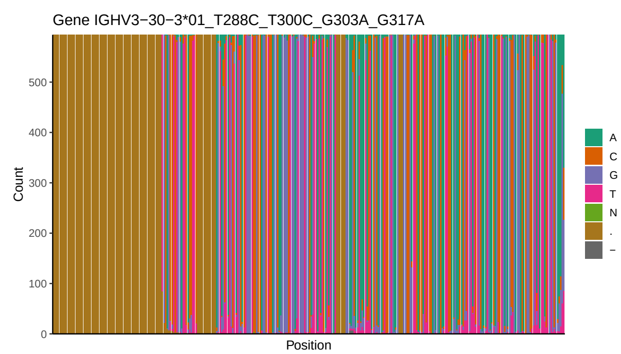


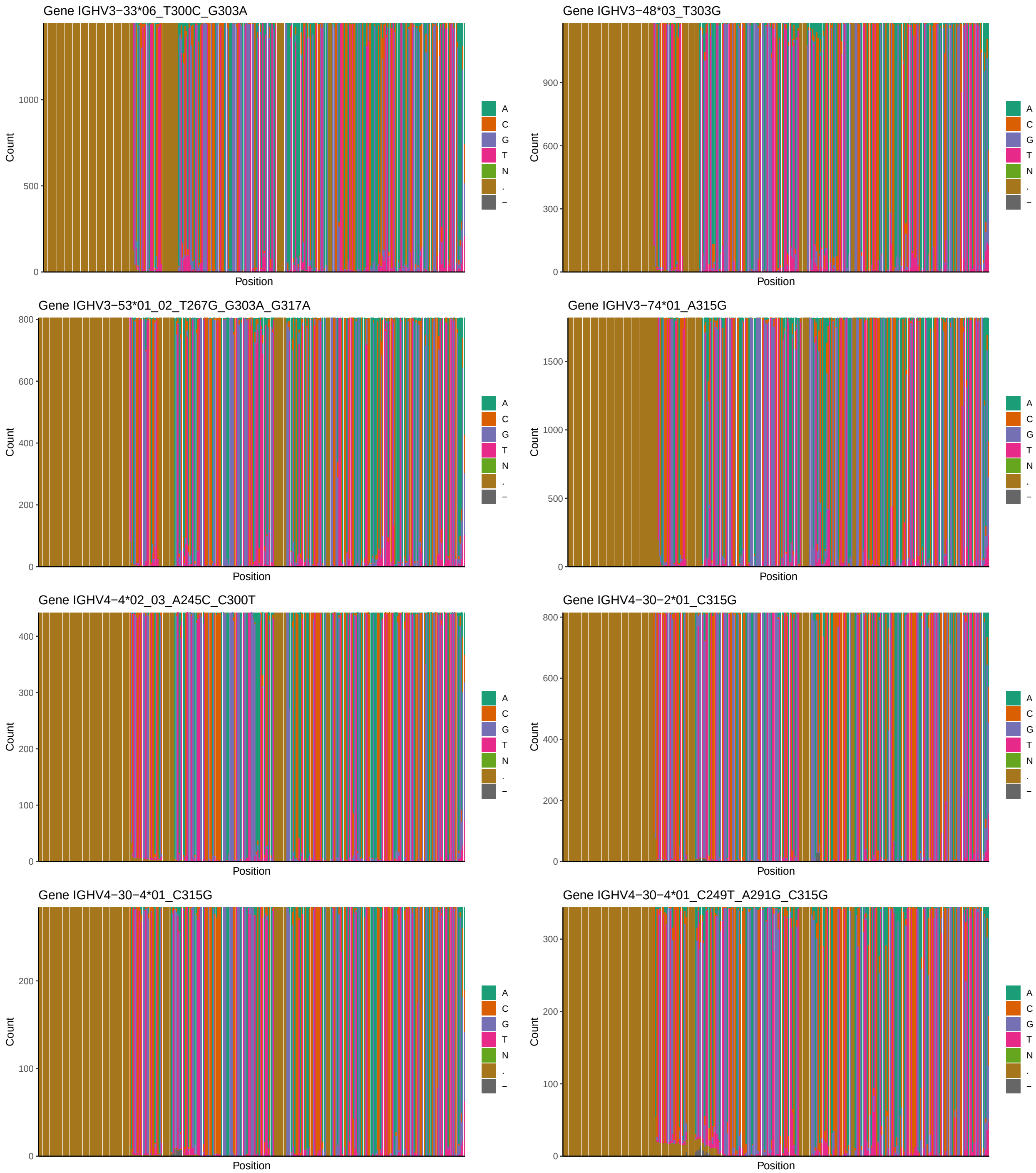


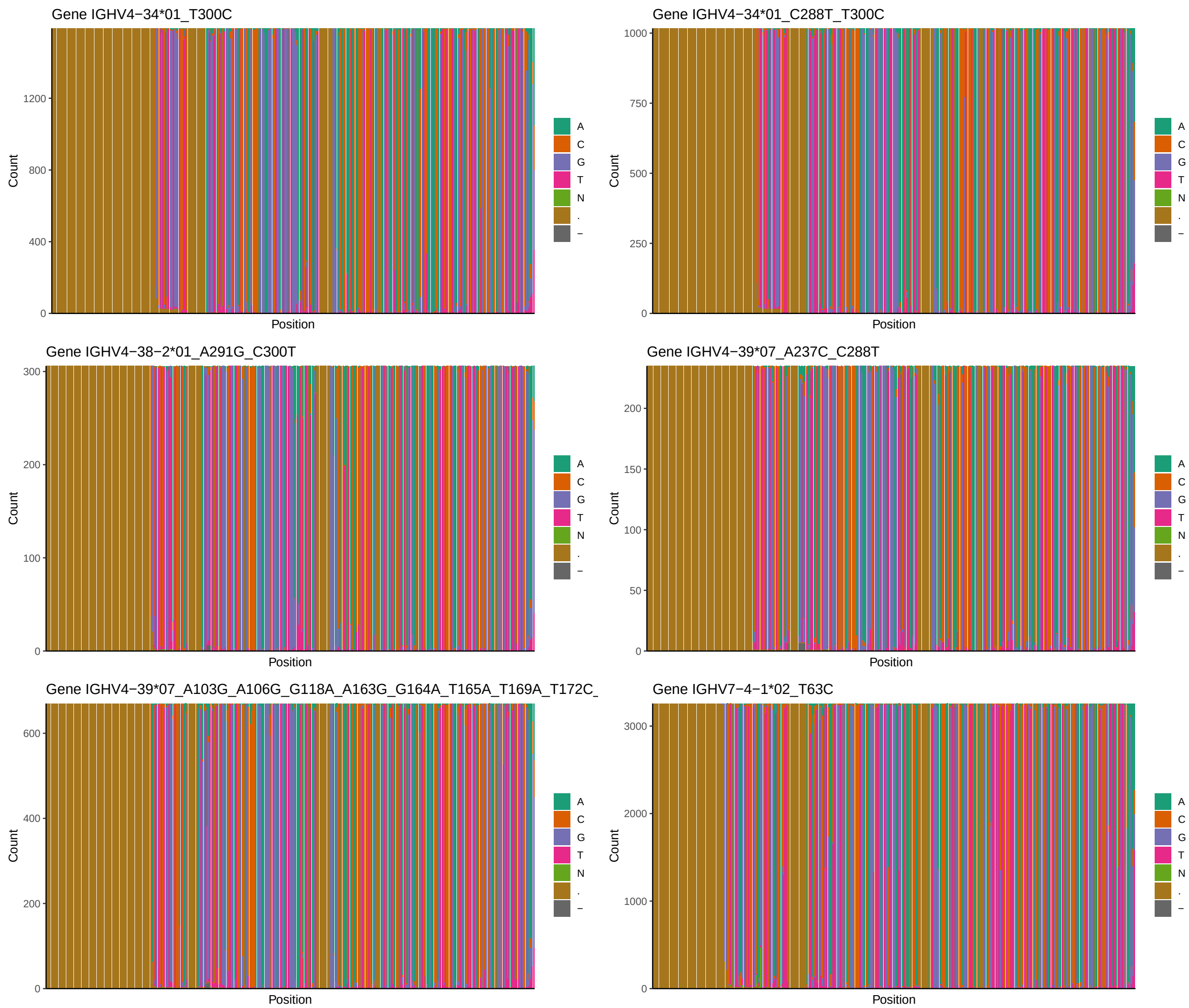




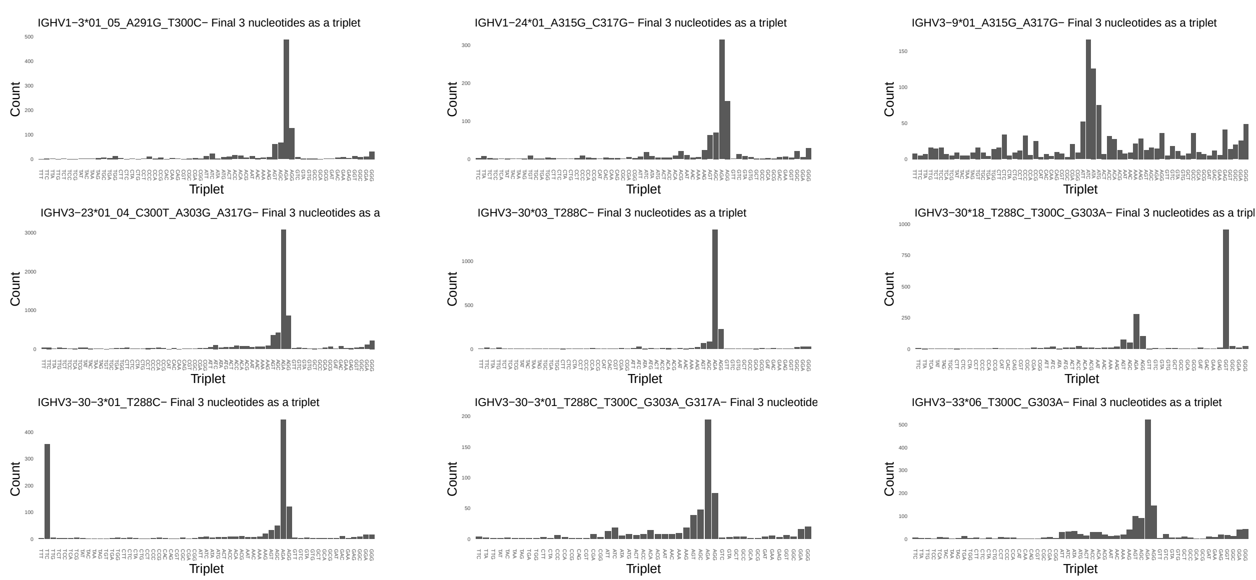


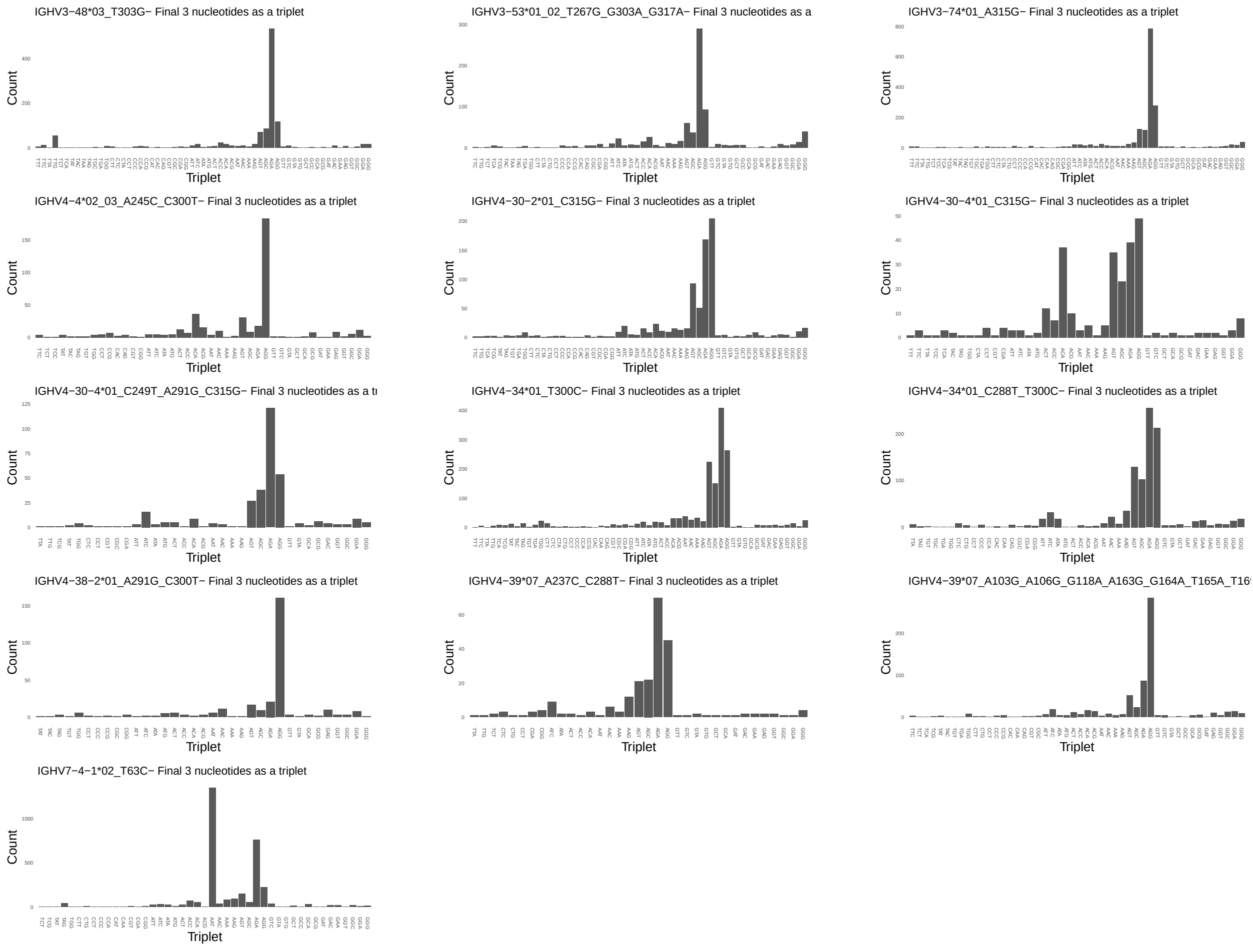



Gene IGHV3−33*06_T300C_G303A
Count
0
500
1000
Position
A
C
G
T
N
.
−
Gene IGHV3−48*03_T303G
Count
0
300
600
900
Position
Gene IGHV3−53*01_02_T267G_G303A_G317A
Count
0
200
400
600
800
Position
Gene IGHV3−74*01_A315G
Count
0
500
1000
1500
Position
Gene IGHV4−4*02_03_A245C_C300T
Count
0
100
200
300
400
Position
Gene IGHV4−30−2*01_C315G
Count
0
200
400
600
800
Position
Gene IGHV4−30−4*01_C315G
Count
0
100
200
Position
Gene IGHV4−30−4*01_C249T_A291G_C315G
Count
0
100
200
300
Position

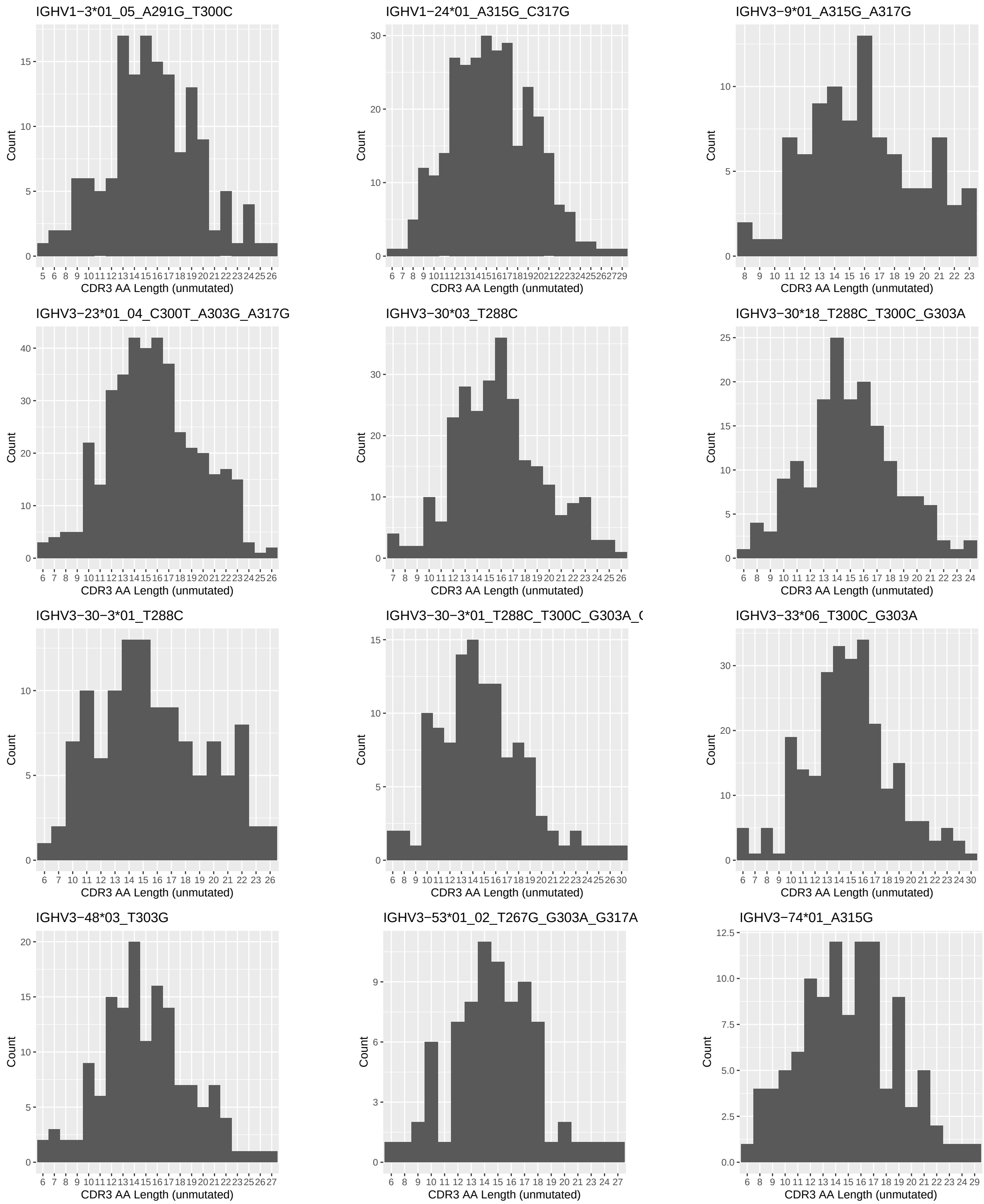


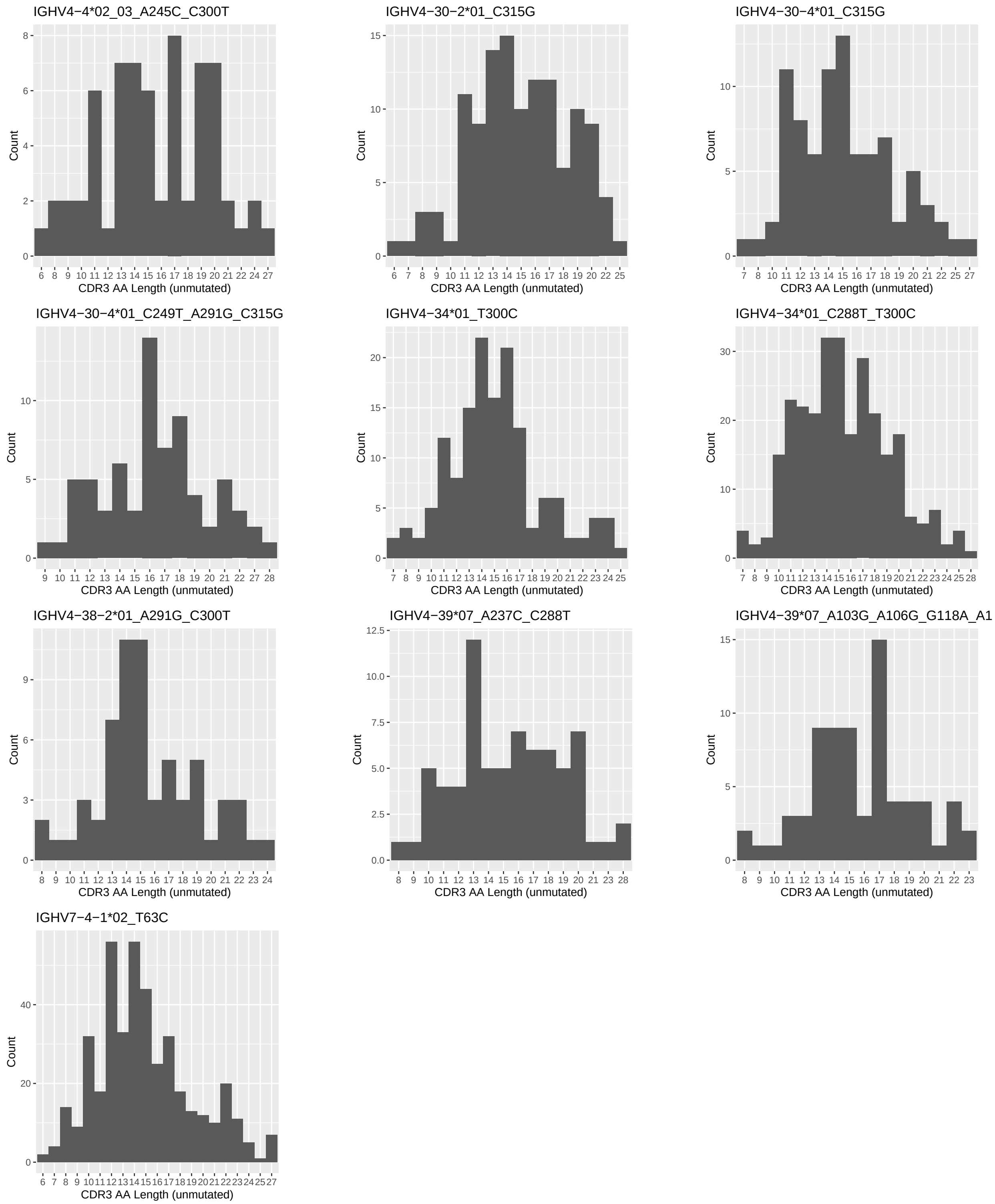
1.4 Final three nucleotides: frequency of each observed triplet



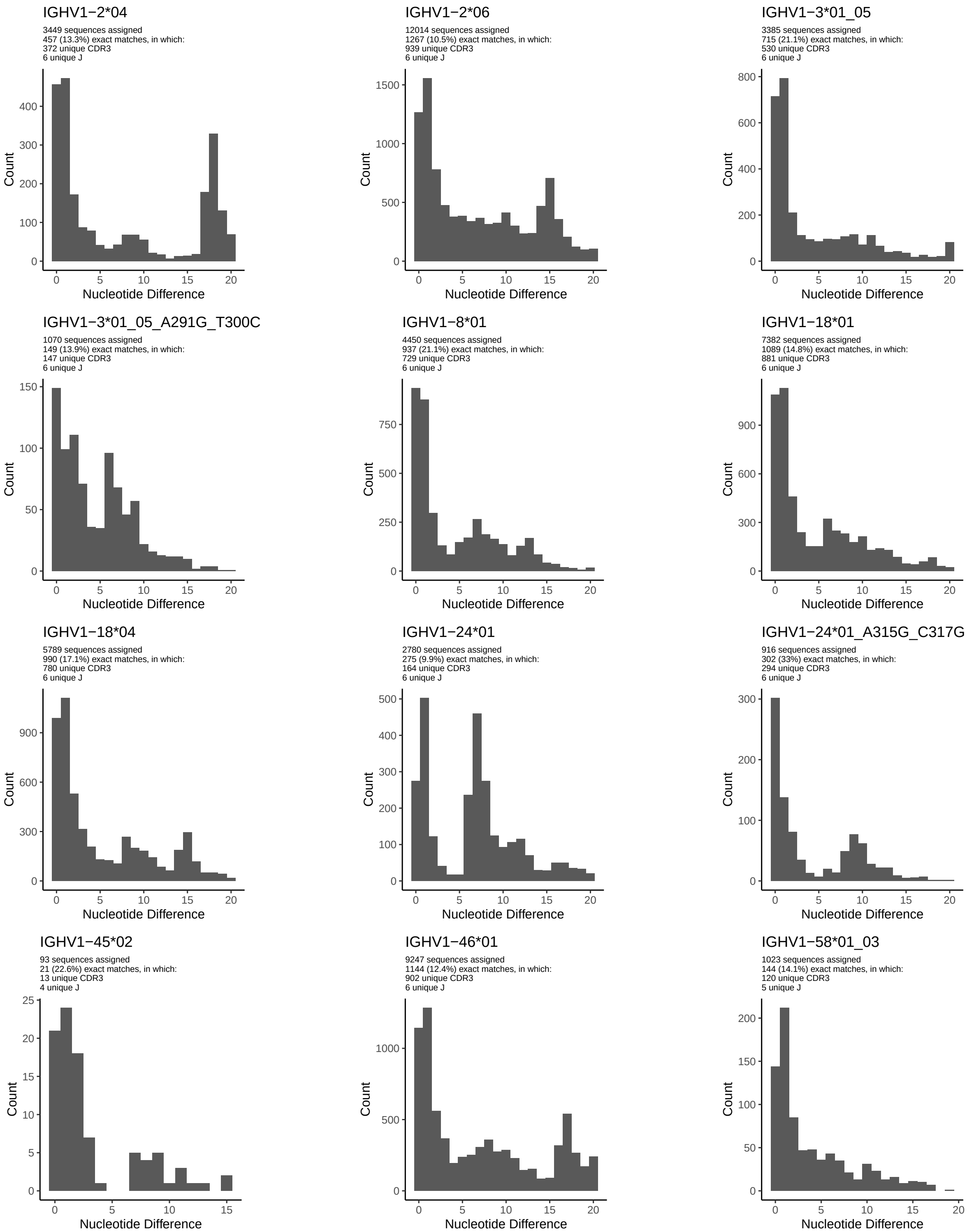

IGHV3−48*03_T303G− Final 3 nucleotides as a triplet
IGHV3−53*01_02_T267G_G303A_G317A− Final 3 nucleotides as a
IGHV3−74*01_A315G− Final 3 nucleotides as a triplet
IGHV4−4*02_03_A245C_C300T− Final 3 nucleotides as a triplet
IGHV4−30−2*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−4*01_C315G− Final 3 nucleotides as a triplet
IGHV4−30−4*01_C249T_A291G_C315G− Final 3 nucleotides as a tr
IGHV4−34*01_T300C− Final 3 nucleotides as a triplet
IGHV4−34*01_C288T_T300C− Final 3 nucleotides as a triplet
IGHV4−38−2*01_A291G_C300T− Final 3 nucleotides as a triplet
IGHV4−39*07_A237C_C288T− Final 3 nucleotides as a triplet
IGHV4−39*07_A103G_A106G_G118A_A163G_G164A_T165A_T16
IGHV7−4−1*02_T63C− Final 3 nucleotides as a triplet
Count
Triplet

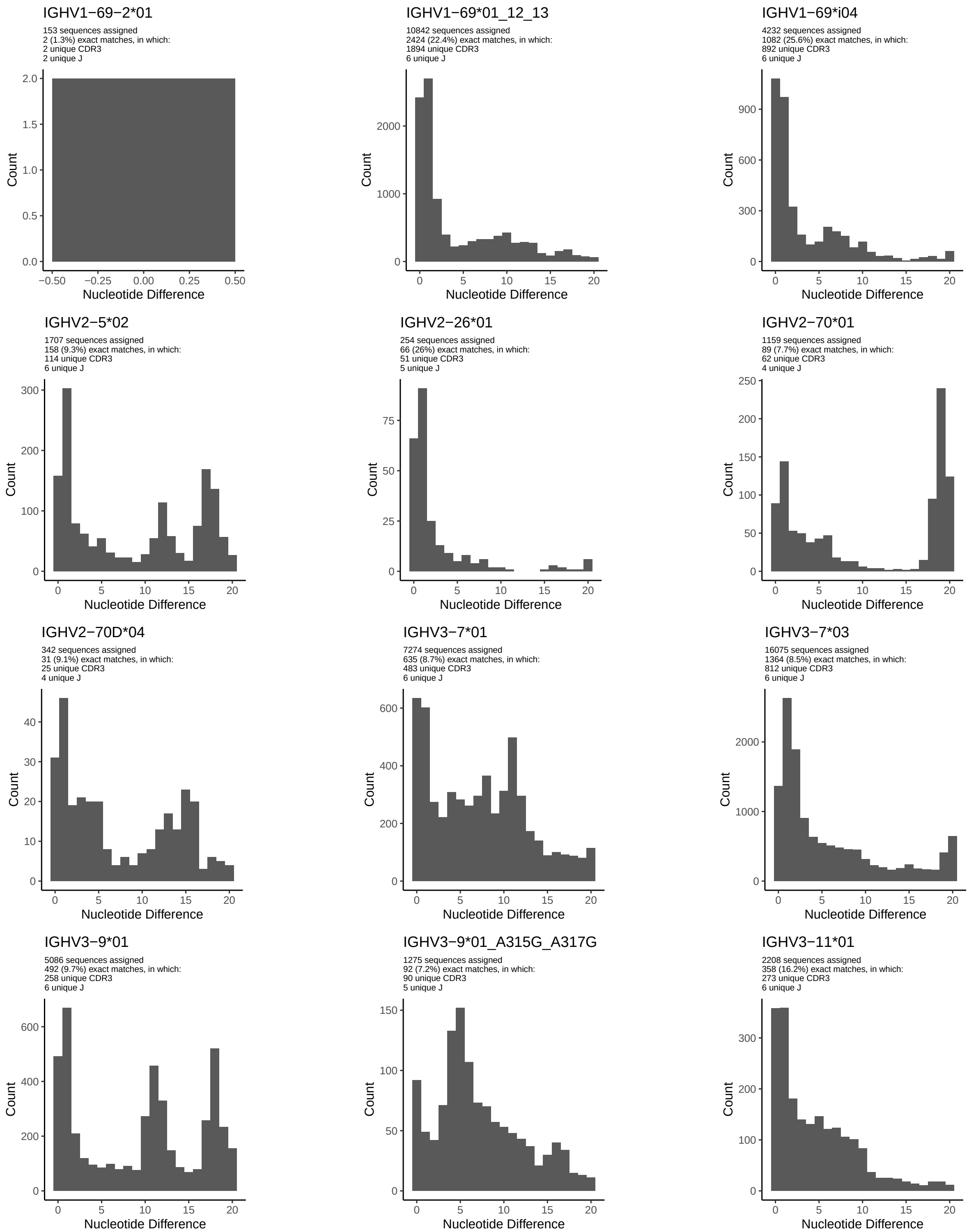
1.5 CDR3 length distribution, in assignments to novel alleles

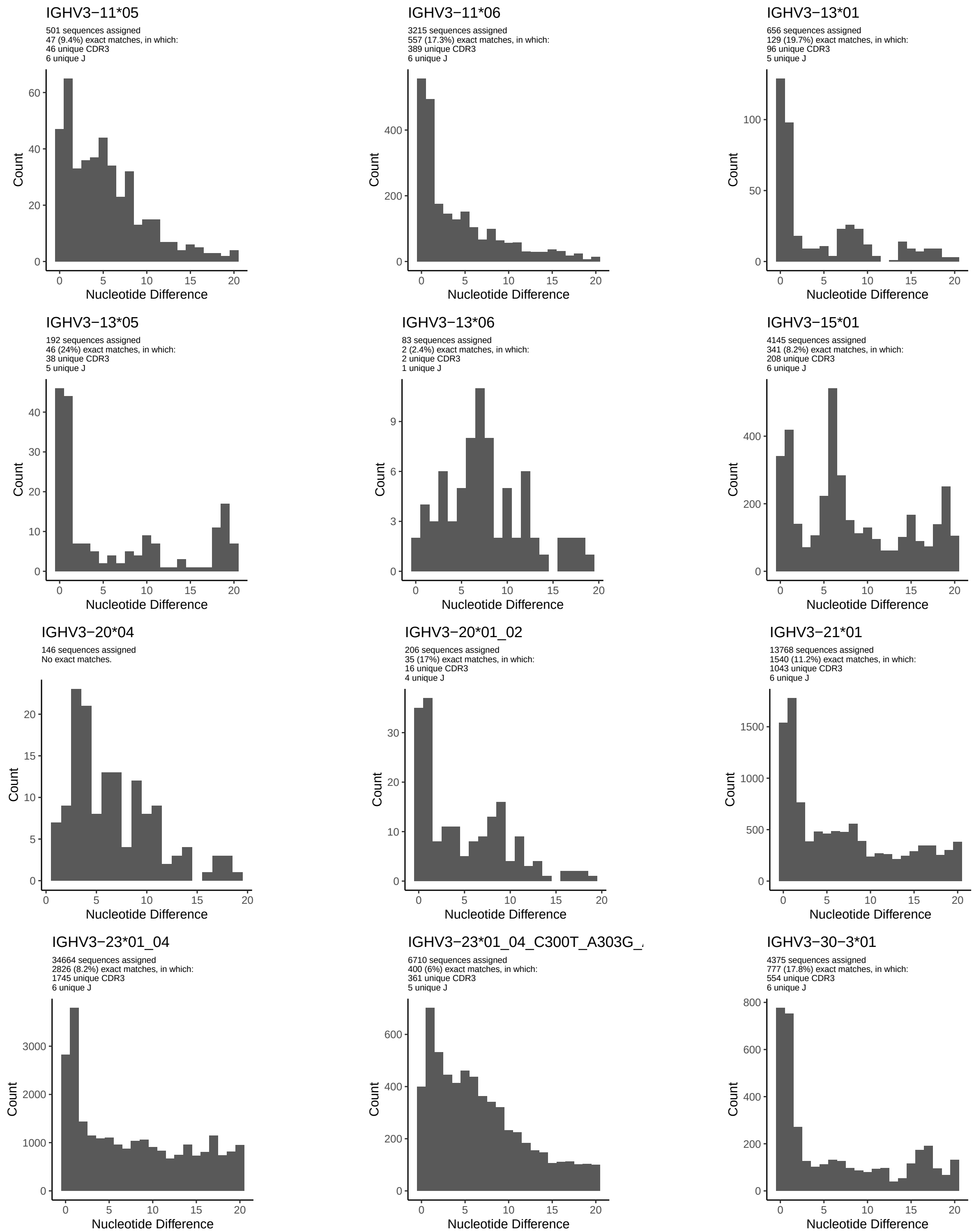



IGHV4−4*02_03_A245C_C300T
IGHV4−30−2*01_C315G
IGHV4−30−4*01_C315G
IGHV4−30−4*01_C249T_A291G_C315G
IGHV4−34*01_T300C
IGHV4−34*01_C288T_T300C
IGHV4−38−2*01_A291G_C300T
IGHV4−39*07_A237C_C288T
IGHV4−39*07_A103G_A106G_G118A_A1
IGHV7−4−1*02_T63C
Count
CDR3 AA Length (unmutated)

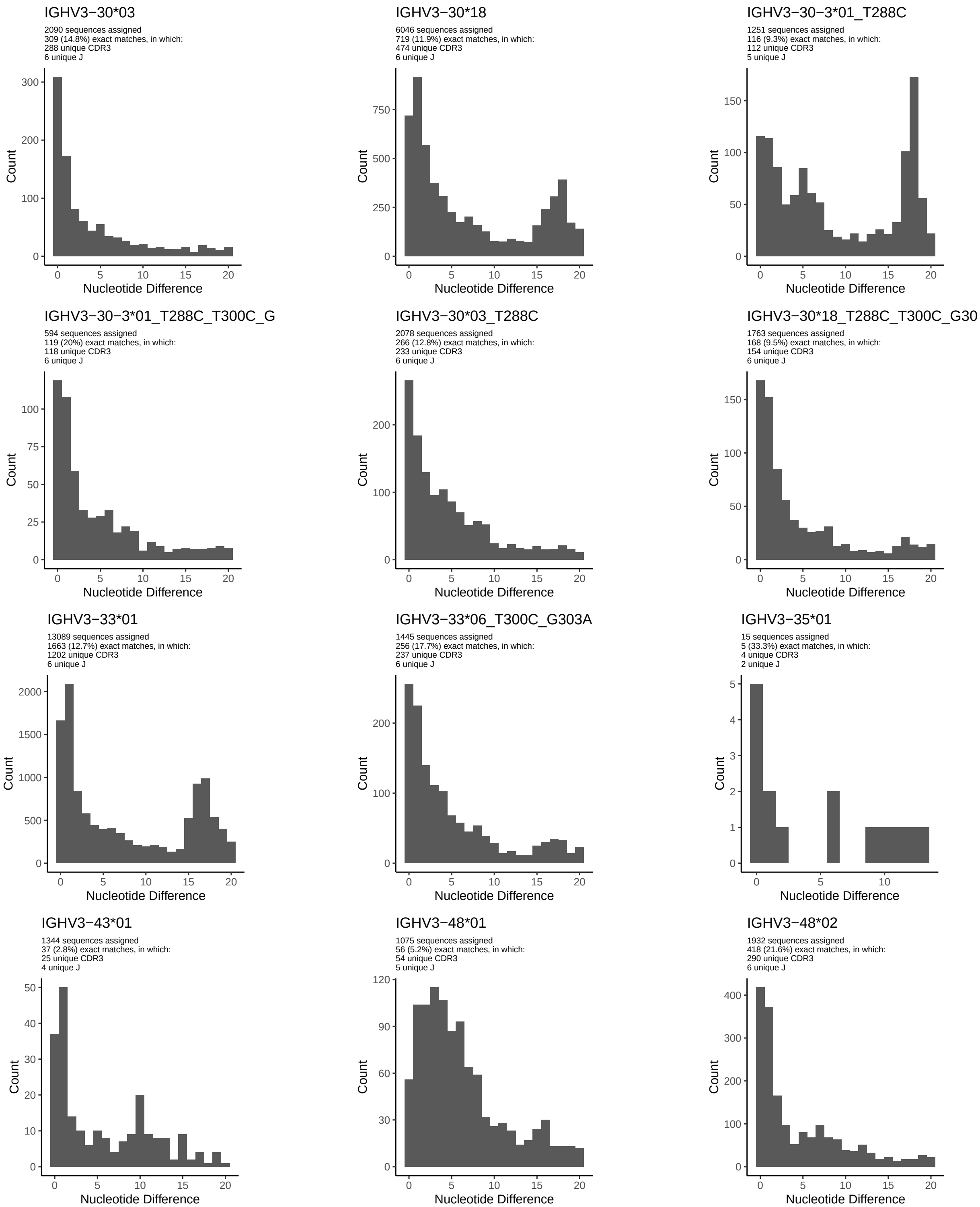
2 Variation from germline, in assignments to each allele

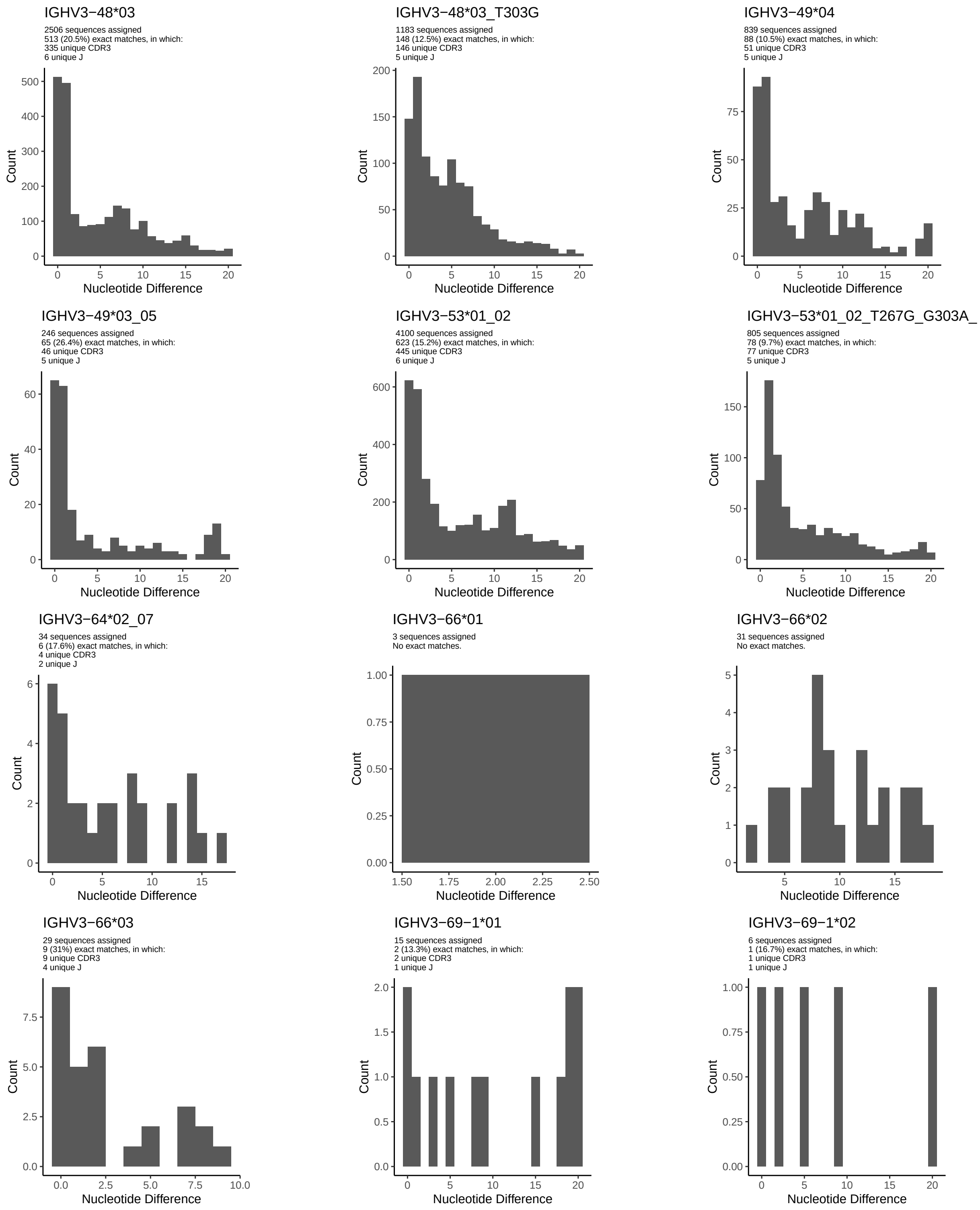


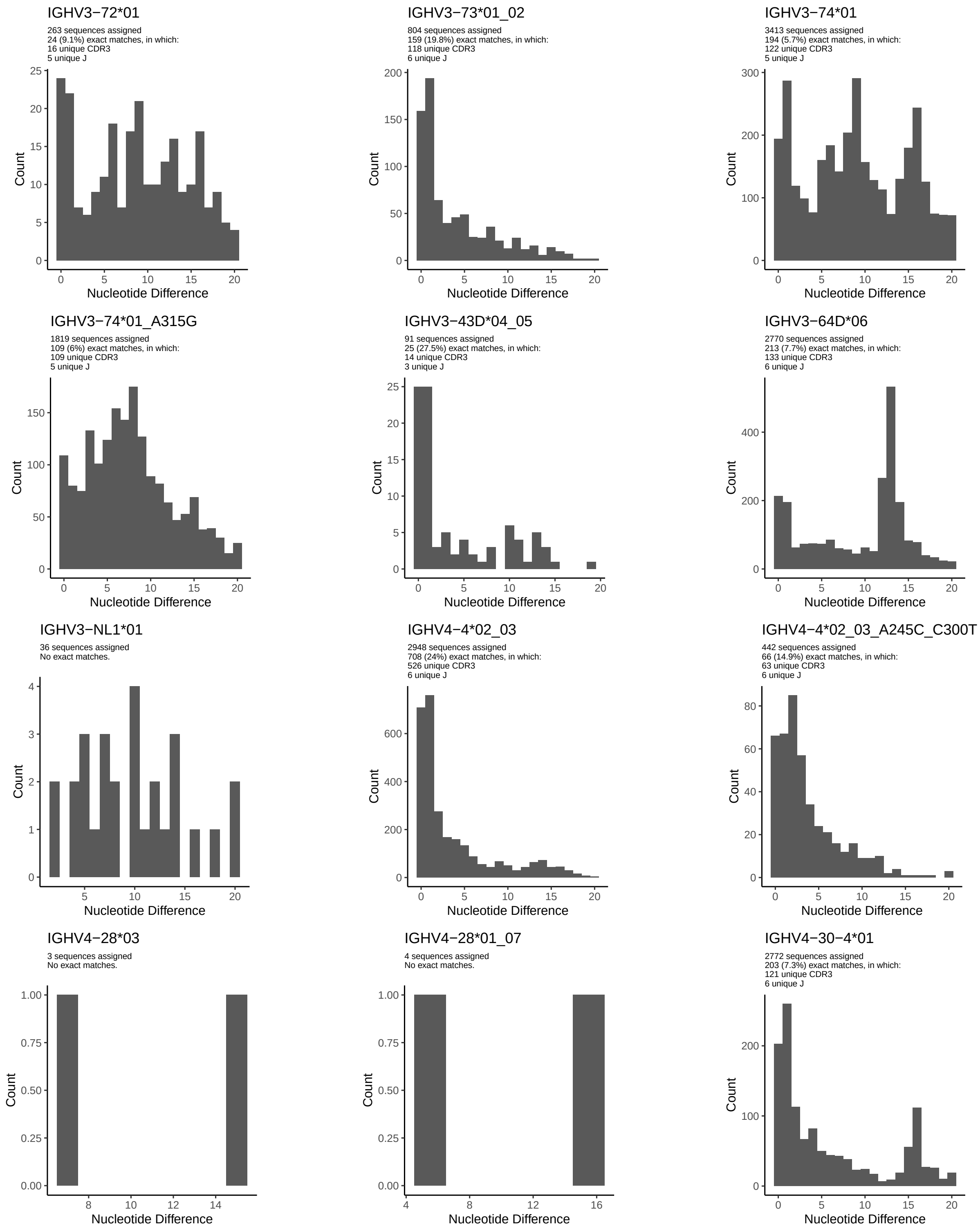

IGHV1−69−2*01
153 sequences assigned
2 (1.3%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV1−69*01_12_13
10842 sequences assigned
2424 (22.4%) exact matches, in which:
1894 unique CDR3
6 unique J
IGHV1−69*i04
4232 sequences assigned
1082 (25.6%) exact matches, in which:
892 unique CDR3
6 unique J
IGHV2−5*02
1707 sequences assigned
158 (9.3%) exact matches, in which:
114 unique CDR3
6 unique J
IGHV2−26*01
254 sequences assigned
66 (26%) exact matches, in which:
51 unique CDR3
5 unique J
IGHV2−70*01
1159 sequences assigned
89 (7.7%) exact matches, in which:
62 unique CDR3
4 unique J
IGHV2−70D*04
342 sequences assigned
31 (9.1%) exact matches, in which:
25 unique CDR3
4 unique J
IGHV3−7*01
7274 sequences assigned
635 (8.7%) exact matches, in which:
483 unique CDR3
6 unique J
IGHV3−7*03
16075 sequences assigned
1364 (8.5%) exact matches, in which:
812 unique CDR3
6 unique J
IGHV3−9*01
5086 sequences assigned
492 (9.7%) exact matches, in which:
258 unique CDR3
6 unique J
IGHV3−9*01_A315G_A317G
1275 sequences assigned
92 (7.2%) exact matches, in which:
90 unique CDR3
5 unique J
IGHV3−11*01
2208 sequences assigned
358 (16.2%) exact matches, in which:
273 unique CDR3
6 unique J
Count
Nucleotide Difference

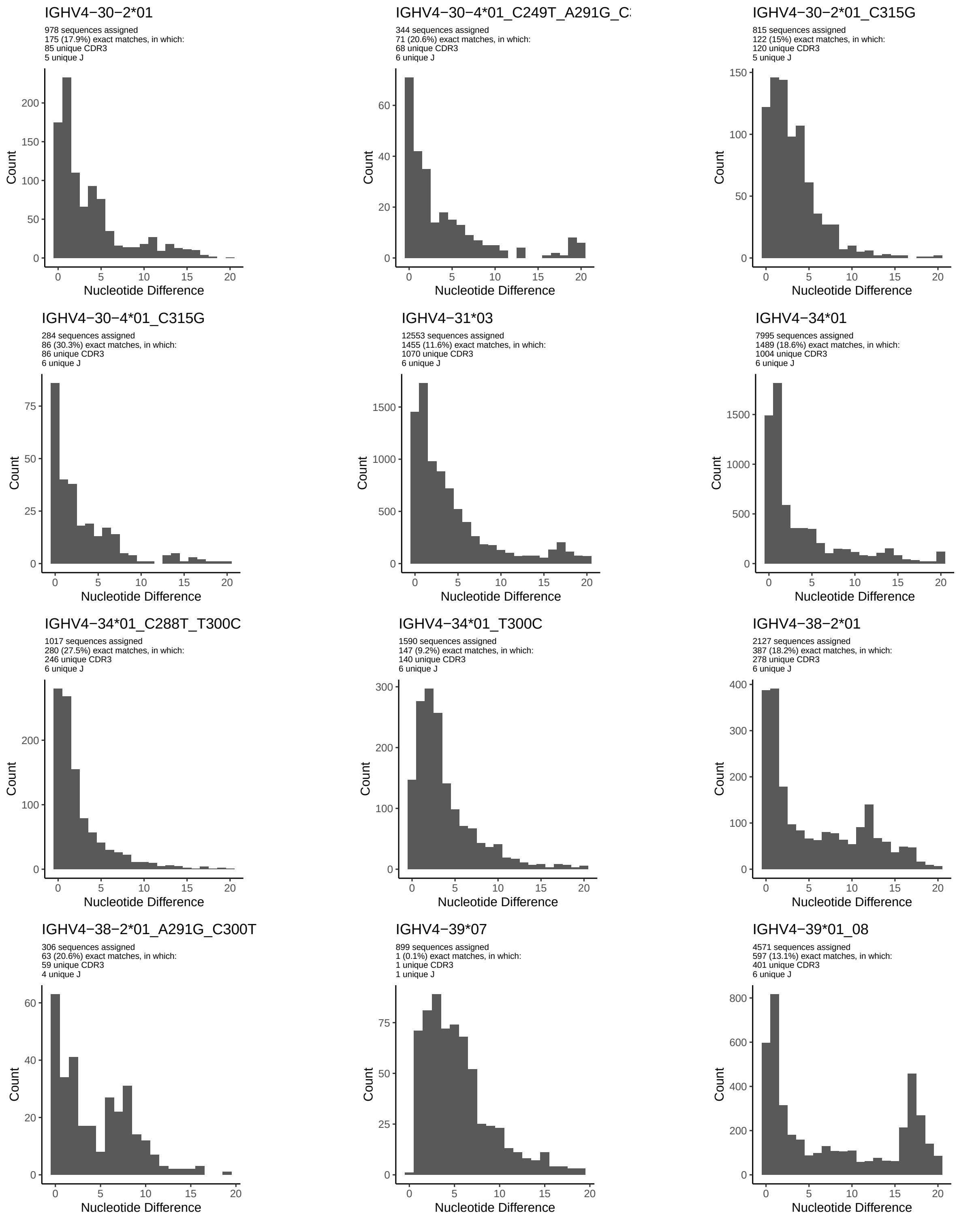


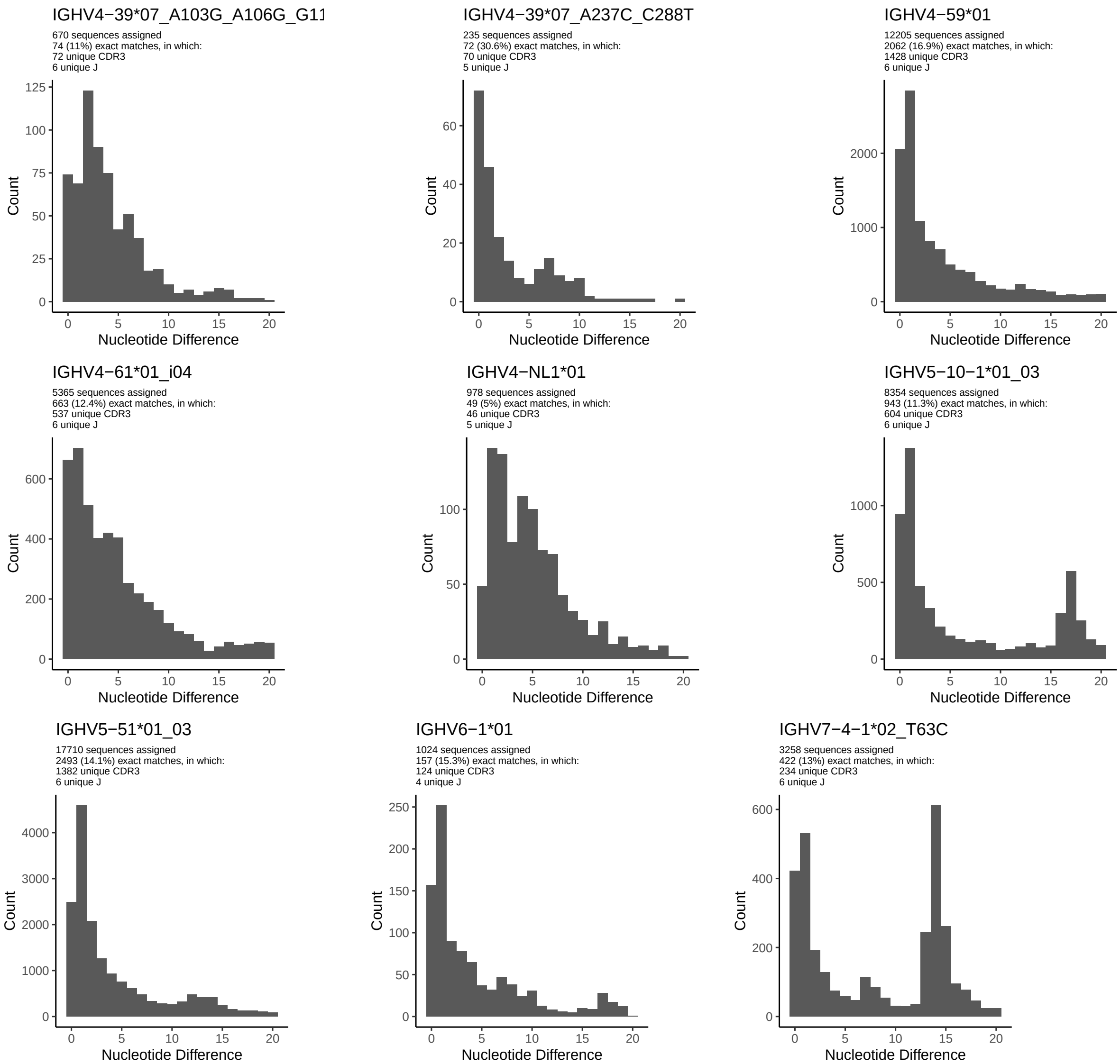
IGHV3−11*05
501 sequences assigned
47 (9.4%) exact matches, in which:
46 unique CDR3
6 unique J
IGHV3−11*06
3215 sequences assigned
557 (17.3%) exact matches, in which:
389 unique CDR3
6 unique J
IGHV3−13*01
656 sequences assigned
129 (19.7%) exact matches, in which:
96 unique CDR3
5 unique J
IGHV3−13*05
192 sequences assigned
46 (24%) exact matches, in which:
38 unique CDR3
5 unique J
IGHV3−13*06
83 sequences assigned
2 (2.4%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV3−15*01
4145 sequences assigned
341 (8.2%) exact matches, in which:
208 unique CDR3
6 unique J
IGHV3−20*04
146 sequences assigned
No exact matches.
IGHV3−20*01_02
206 sequences assigned
35 (17%) exact matches, in which:
16 unique CDR3
4 unique J
IGHV3−21*01
13768 sequences assigned
1540 (11.2%) exact matches, in which:
1043 unique CDR3
6 unique J
IGHV3−23*01_04
34664 sequences assigned
2826 (8.2%) exact matches, in which:
1745 unique CDR3
6 unique J
IGHV3−23*01_04_C300T_A303G_
6710 sequences assigned
400 (6%) exact matches, in which:
361 unique CDR3
5 unique J
IGHV3−30−3*01
4375 sequences assigned
777 (17.8%) exact matches, in which:
554 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−30*03
2090 sequences assigned
309 (14.8%) exact matches, in which:
288 unique CDR3
6 unique J
IGHV3−30*18
6046 sequences assigned
719 (11.9%) exact matches, in which:
474 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
1251 sequences assigned
116 (9.3%) exact matches, in which:
112 unique CDR3
5 unique J
IGHV3−30−3*01_T288C_T300C_G
594 sequences assigned
119 (20%) exact matches, in which:
118 unique CDR3
6 unique J
IGHV3−30*03_T288C
2078 sequences assigned
266 (12.8%) exact matches, in which:
233 unique CDR3
6 unique J
IGHV3−30*18_T288C_T300C_G30
1763 sequences assigned
168 (9.5%) exact matches, in which:
154 unique CDR3
6 unique J
IGHV3−33*01
13089 sequences assigned
1663 (12.7%) exact matches, in which:
1202 unique CDR3
6 unique J
IGHV3−33*06_T300C_G303A
1445 sequences assigned
256 (17.7%) exact matches, in which:
237 unique CDR3
6 unique J
IGHV3−35*01
15 sequences assigned
5 (33.3%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV3−43*01
1344 sequences assigned
37 (2.8%) exact matches, in which:
25 unique CDR3
4 unique J
IGHV3−48*01
1075 sequences assigned
56 (5.2%) exact matches, in which:
54 unique CDR3
5 unique J
IGHV3−48*02
1932 sequences assigned
418 (21.6%) exact matches, in which:
290 unique CDR3
6 unique J
Count
Nucleotide Difference

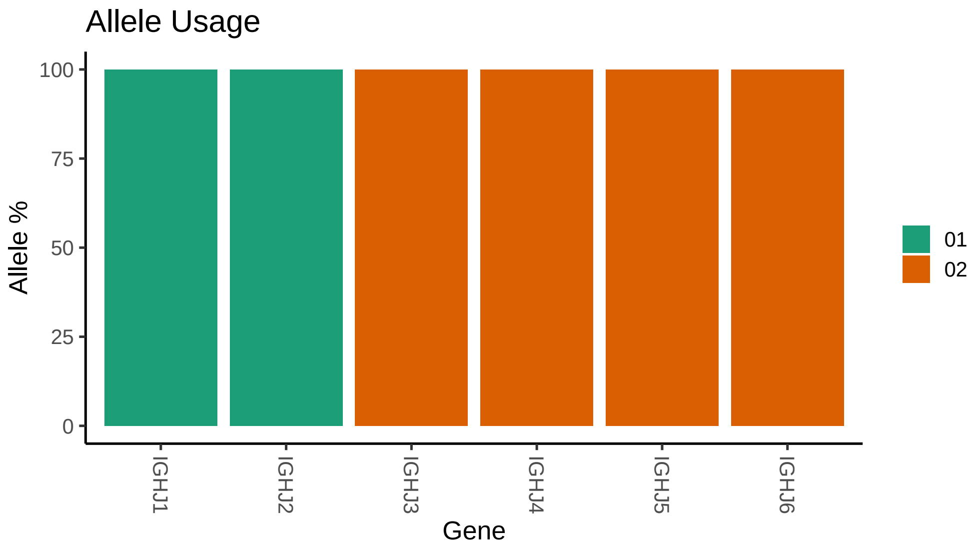

IGHV3−48*03
2506 sequences assigned
513 (20.5%) exact matches, in which:
335 unique CDR3
6 unique J
IGHV3−48*03_T303G
1183 sequences assigned
148 (12.5%) exact matches, in which:
146 unique CDR3
5 unique J
IGHV3−49*04
839 sequences assigned
88 (10.5%) exact matches, in which:
51 unique CDR3
5 unique J
IGHV3−49*03_05
246 sequences assigned
65 (26.4%) exact matches, in which:
46 unique CDR3
5 unique J
IGHV3−53*01_02
4100 sequences assigned
623 (15.2%) exact matches, in which:
445 unique CDR3
6 unique J
IGHV3−53*01_02_T267G_G303A_
805 sequences assigned
78 (9.7%) exact matches, in which:
77 unique CDR3
5 unique J
IGHV3−64*02_07
34 sequences assigned
6 (17.6%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV3−66*01
3 sequences assigned
No exact matches.
IGHV3−66*02
31 sequences assigned
No exact matches.
IGHV3−66*03
29 sequences assigned
9 (31%) exact matches, in which:
9 unique CDR3
4 unique J
IGHV3−69−1*01
15 sequences assigned
2 (13.3%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV3−69−1*02
6 sequences assigned
1 (16.7%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference


IGHV3−72*01
263 sequences assigned
24 (9.1%) exact matches, in which:
16 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−73*01_02
804 sequences assigned
159 (19.8%) exact matches, in which:
118 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−74*01
3413 sequences assigned
194 (5.7%) exact matches, in which:
122 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−74*01_A315G
1819 sequences assigned
109 (6%) exact matches, in which:
109 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV3−43D*04_05
91 sequences assigned
25 (27.5%) exact matches, in which:
14 unique CDR3
3 unique J
Count
Nucleotide Difference
IGHV3−64D*06
2770 sequences assigned
213 (7.7%) exact matches, in which:
133 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−NL1*01
36 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV4−4*02_03
2948 sequences assigned
708 (24%) exact matches, in which:
526 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−4*02_03_A245C_C300T
442 sequences assigned
66 (14.9%) exact matches, in which:
63 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−28*03
3 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV4−28*01_07
4 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV4−30−4*01
2772 sequences assigned
203 (7.3%) exact matches, in which:
121 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−30−2*01
978 sequences assigned
175 (17.9%) exact matches, in which:
85 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV4−30−4*01_C249T_A291G_C
344 sequences assigned
71 (20.6%) exact matches, in which:
68 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−30−2*01_C315G
815 sequences assigned
122 (15%) exact matches, in which:
120 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV4−30−4*01_C315G
284 sequences assigned
86 (30.3%) exact matches, in which:
86 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−31*03
12553 sequences assigned
1455 (11.6%) exact matches, in which:
1070 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−34*01
7995 sequences assigned
1489 (18.6%) exact matches, in which:
1004 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−34*01_C288T_T300C
1017 sequences assigned
280 (27.5%) exact matches, in which:
246 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−34*01_T300C
1590 sequences assigned
147 (9.2%) exact matches, in which:
140 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−38−2*01
2127 sequences assigned
387 (18.2%) exact matches, in which:
278 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV4−38−2*01_A291G_C300T
306 sequences assigned
63 (20.6%) exact matches, in which:
59 unique CDR3
4 unique J
Count
Nucleotide Difference
IGHV4−39*07
899 sequences assigned
1 (0.1%) exact matches, in which:
1 unique CDR3
1 unique J
Count
Nucleotide Difference
IGHV4−39*01_08
4571 sequences assigned
597 (13.1%) exact matches, in which:
401 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−39*07_A103G_A106G_G11
670 sequences assigned
74 (11%) exact matches, in which:
72 unique CDR3
6 unique J
IGHV4−39*07_A237C_C288T
235 sequences assigned
72 (30.6%) exact matches, in which:
70 unique CDR3
5 unique J
IGHV4−59*01
12205 sequences assigned
2062 (16.9%) exact matches, in which:
1428 unique CDR3
6 unique J
IGHV4−61*01_i04
5365 sequences assigned
663 (12.4%) exact matches, in which:
537 unique CDR3
6 unique J
IGHV4−NL1*01
978 sequences assigned
49 (5%) exact matches, in which:
46 unique CDR3
5 unique J
IGHV5−10−1*01_03
8354 sequences assigned
943 (11.3%) exact matches, in which:
604 unique CDR3
6 unique J
IGHV5−51*01_03
17710 sequences assigned
2493 (14.1%) exact matches, in which:
1382 unique CDR3
6 unique J
IGHV6−1*01
1024 sequences assigned
157 (15.3%) exact matches, in which:
124 unique CDR3
4 unique J
IGHV7−4−1*02_T63C
3258 sequences assigned
422 (13%) exact matches, in which:
234 unique CDR3
6 unique J
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 005/M64 005/M64 005/M64 005/68/c47e9ca8f3b640eff
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 005/M64 005/M64 005/M64 005/68/c47e9ca8f
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 005/M64 005/M64 005/M64 005/68/c47e9ca8f3b640e
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 9 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 9 01_A315G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 18: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 18_T288C_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C_T300C_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 48 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 48 03_T303G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 53 01_02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 53 01_02_T267G_G303A_G317A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C249T_A291G_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 01_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 39 07: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 07_A237C_C288T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 07_A103G_A106G_G118A_A163G_G164A_T165A_T169A_T172C_G189A_T19
```